



THE KING'S SCHOOL

2005 Higher School Certificate Trial Examination

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1-8
- All questions are of equal value



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Mathematics Extension 2

Question	Complex Numbers	Functions	Integration	Conics	Mechanics	Harder Extension 1	Total
1		(d)	(a), (b), (c)				15
2	(b), (c), (d), (e)		(a)				15
3		(a), (b)(i)(ii)(iii)	(b)(iv)(v)				15
4		(b)	(a)	(c)			15
5			(b)		(a)		15
6		(a)				(b)	15
7						(a), (b)	15
8	(a)		(b)				15
Marks	20	24	37	9	9	21	120

Total marks – 120
Attempt Questions 1-8
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

Question 1 (15 marks) Use a SEPARATE writing booklet.

(a) (i) Express $\frac{2}{1-x^2}$ in partial fractions. 2

(ii) Show that $\int_0^{\frac{1}{2}} \frac{2}{1-x^2} dx = \ln\left(\frac{5}{3}\right)$ 2

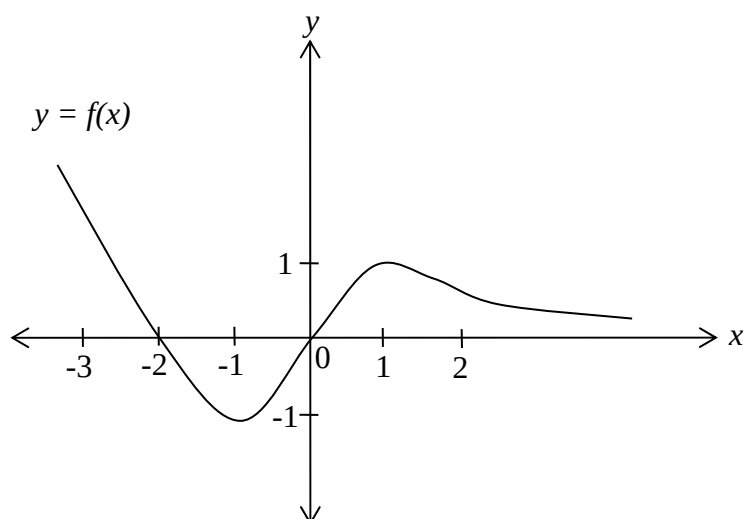
(iii) Evaluate $\int_0^{\frac{1}{2}} \frac{2x}{1-x^4} dx$ 2

(b) Evaluate $\int_0^{\frac{\pi}{4}} \frac{2}{1 + \sin 2x + \cos 2x} dx$ 3

(c) Use completion of square to prove that $\int_0^1 \frac{4}{4x^2 + 4x + 5} dx = \tan^{-1}\left(\frac{4}{7}\right)$ 3

Question 1 is continued on the next page

(d)



On separate diagrams, sketch the graphs of:

(i) $y = \ln f(x)$

2

(ii) $y = e^{\ln f(x)}$

1

End of Question 1

- (a) (i) Use integration by parts to show that

$$\int_0^1 (x-1) f'(x) dx = f(0) - \int_0^1 f(x) dx \quad 2$$

(ii) Hence, or otherwise, evaluate $\int_0^1 \frac{x-1}{(x+1)^2} dx$ 2

- (b) Let $z = x + iy$, x, y real, where $\arg z = \frac{3\pi}{5}$

- (i) Sketch the locus of z 1

- (ii) Find $\arg(-z)$ 1

- (c) Sketch the region in the complex plane where $|z - i| \leq |z + 1|$ 2

- (d) $z = x + iy$, x, y real, is a complex number such that

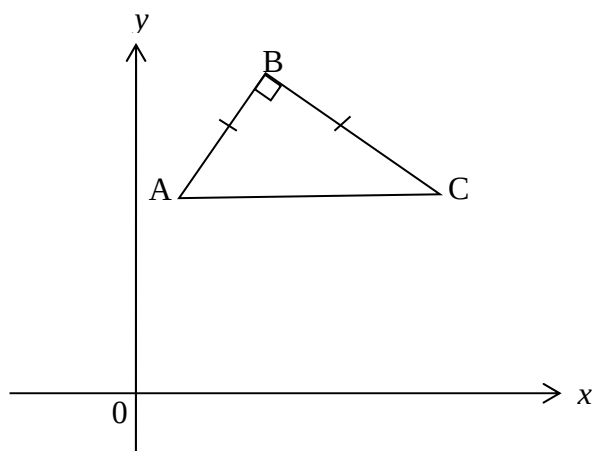
$$(z + \bar{z})^2 + (z - \bar{z})^2 = 4$$

- (i) Find the cartesian locus of z 2

- (ii) Sketch the locus of z in the complex plane showing any features necessary to indicate your diagram clearly. 2

Question 2 is continued on the next page

(e)



In the Argand diagram, $\triangle ABC$ is right-angled at B and isosceles.

A, B, C represent the complex numbers a , b , c respectively.

- (i) Find the complex number $\overrightarrow{BA}$ in terms of a and b . 1
- (ii) Prove that $c = ai + b(1 - i)$ 2

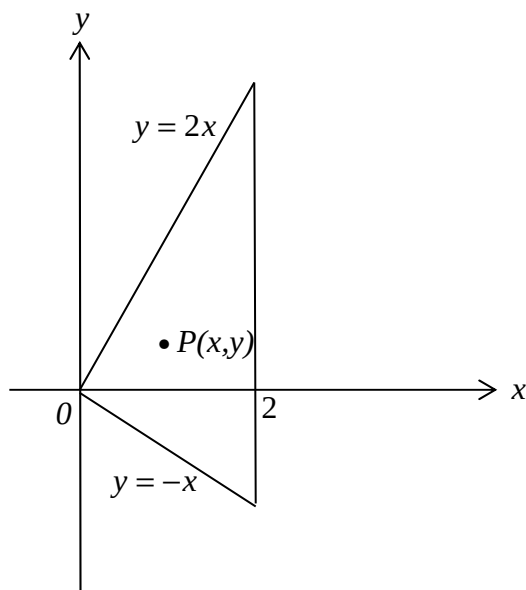
End of Question 2

- (a) (i) Sketch the parabola $y = \frac{1+x^2}{2}$ and use it to sketch the curve $y = \frac{2}{1+x^2}$ on the same diagram. 2
- (ii) Hence, or otherwise, find the range of the function $y = \frac{2}{1+x^2} - 1$ 1
- (b) Consider the function $y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$
- (i) By using (a), or otherwise, find the range of the function. 2
- (ii) Show that $\frac{d}{dx} \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \frac{2x}{(1+x^2)\sqrt{x^2}}$ and give the simplest expressions for the derivative if
 $(\alpha) \ x > 0$ and $(\beta) \ x < 0$ 3
- (iii) Sketch the curve $y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ 2
- (iv) The region bounded by $y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ and the line $y = \frac{\pi}{2}$ is revolved about the y axis.
 Show that the volume of the solid of revolution is given by

$$V = \pi \int_0^{\frac{\pi}{2}} \frac{1 - \cos y}{1 + \cos y} dy$$
 2
- (v) Find the volume V . 3

End of Question 3

(a)



The base of a solid is the triangular region bounded by the lines $y = 2x$, $y = -x$ and $x = 2$.

At each point $P(x, y)$ in the base the height of the solid is $4x^2 + x$

Find the volume of the solid.

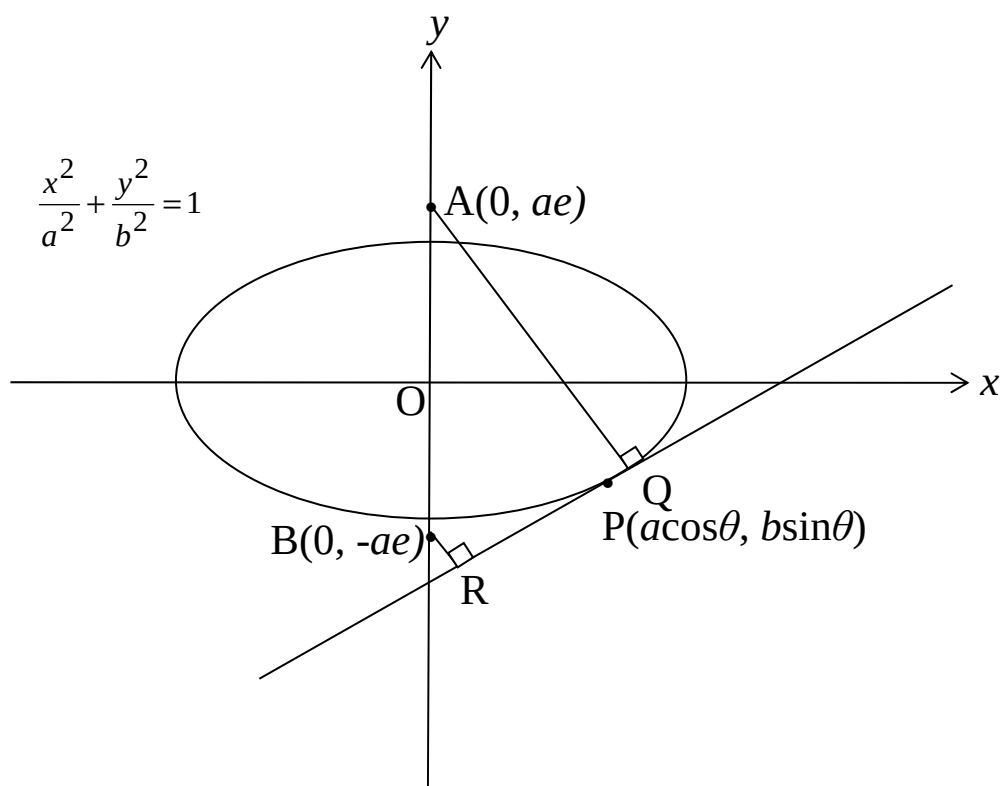
4

(b) If $xy^2 + 1 = x^2$, $y \neq 0$, show that $\frac{dy}{dx} = \frac{1}{y} - \frac{y}{2x}$

2

Question 4 is continued on the next page

(c)



$P(\operatorname{acos}\theta, b\sin\theta)$ is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b > 0$, where e is the eccentricity of the ellipse.

From $A(0, ae)$ and $B(0, -ae)$ perpendiculars are drawn to meet the tangent at $P(\operatorname{acos}\theta, b\sin\theta)$ at Q and R , respectively.

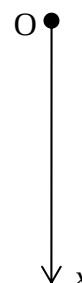
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|-------|--|---|
| (i) | Prove that the equation of the tangent at P is $\frac{\cos\theta}{a}x + \frac{\sin\theta}{b}y = 1$ | 3 |
| (ii) | Hence, or otherwise, show that the line $x \cos \alpha + y \sin \alpha = k$ is a tangent to the ellipse if $a^2 \cos^2 \alpha + b^2 \sin^2 \alpha = k^2$ | 2 |
| (iii) | Hence, or otherwise, prove that $AQ^2 + BR^2 = 2a^2$ | 4 |

End of Question 4

- (a) A particle of mass m moving with speed v experiences air resistance mkv^2 , where k is a positive constant. g is the constant acceleration due to gravity.

- (i) The particle of mass m falls from rest from a point O.

Taking the positive x axis as vertically downward, show that $\ddot{x} = k(V^2 - v^2)$, where V is the terminal speed.



2

- (ii) Another particle of mass m is projected vertically upward from ground level with a speed V^2 , where V is the terminal speed as in (i).

Prove that the particle will reach a maximum height of $\frac{1}{2k} \ln(1 + V^2)$

3

- (iii) Prove that the particle in (ii) will return to the ground with speed U where $U^{-2} = V^{-2} + V^{-4}$

4

- (b) The ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$ is revolved about the line $x = 4$.

- (i) Use the method of cylindrical shells to show that the volume of the solid of revolution is given by

$$V = 8\sqrt{3} \pi \int_{-2}^2 \sqrt{4-x^2} \, dx - 2\sqrt{3} \pi \int_{-2}^2 x \sqrt{4-x^2} \, dx$$

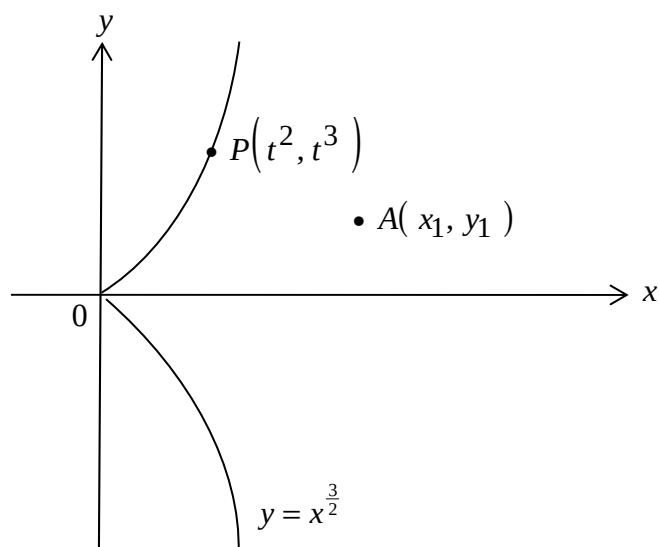
4

- (ii) Prove that the volume $V = 16\sqrt{3} \pi^2$

2

End of Question 5

(a)



$P(t^2, t^3)$ is any point in the curve $y = x^{\frac{3}{2}}$

(i) Show that the equation of the tangent at $P(t^2, t^3)$ is $3tx - 2y - t^3 = 0$ 2

(ii) $A(x_1, y_1)$ is a point not on the curve $y = x^{\frac{3}{2}}$

Deduce that at most three tangents to the curve pass through A . 1

(iii) If the tangents with parameters t_1, t_2, t_3 do pass through $A(x_1, y_1)$, show that

$$(\alpha) \quad t_1^3 + t_2^3 + t_3^3 = -6y_1 \quad \text{2}$$

$$(\beta) \quad (t_1 t_2)^2 + (t_2 t_3)^2 + (t_3 t_1)^2 = 9x_1^2 \quad \text{2}$$

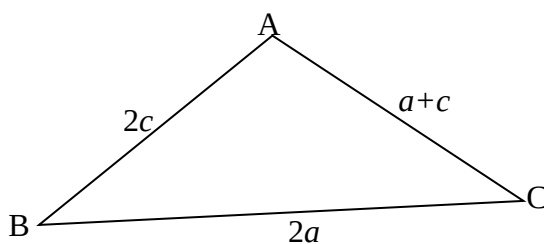
(iv) Find a cubic equation with roots $\frac{1}{t_1}, \frac{1}{t_2}, \frac{1}{t_3}$ 2

Question 6 is continued on the next page

- (b) (i) Given that $\sin(X + Y) + \sin(X - Y) = 2 \sin X \cos Y$, show that

$$\sin A + \sin C = 2 \sin \frac{A+C}{2} \cos \frac{A-C}{2} \quad \mathbf{1}$$

- (ii) Consider $\triangle ABC$ where



- (α) Use the sine rule to show that $\sin A + \sin C = 2 \sin B$ **2**

- (β) Deduce that $\sin \frac{B}{2} = \frac{1}{2} \cos \frac{A-C}{2}$ **3**

End of Question 6

(a) Let $f(n) = (n+1)^3 + (n+2)^3 + \dots + (2n-1)^3 + (2n)^3$, $n = 1, 2, 3, \dots$

(i) Show that $f(n+1) - f(n) = (2n+1)^3 + 7(n+1)^3$ 2

(ii) Show that

$$(2n+1)^3 - \frac{2n+1}{4}(3n+1)(5n+3) = \frac{2n+1}{4}(n+1)^2$$
 1

(iii) Use mathematical induction for integers $n = 1, 2, 3, \dots$ to prove that

$$f(n) = (n+1)^3 + (n+2)^3 + \dots + (2n)^3 = \frac{n^2}{4}(3n+1)(5n+3)$$
 4

(iv) Given that $1^3 + 2^3 + \dots + n^3 = \left[\frac{n}{2}(n+1) \right]^2$, prove that

$$(n+1)^3 + (n+2)^3 + \dots + (2n)^3 = \frac{n^2}{4}(3n+1)(5n+3) \text{ without induction.}$$
 2

(b) (i) Show that $\frac{\binom{n}{k}}{n^k} = \frac{\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{k-1}{n}\right)}{k!}$, $2 \leq k \leq n$ 2

(ii) Deduce that $\frac{\binom{n+1}{k}}{(n+1)^k} > \frac{\binom{n}{k}}{n^k}$, $2 \leq k \leq n$ 2

(iii) Deduce that, if n is a positive integer, $\left(1 + \frac{1}{n+1}\right)^{n+1} > \left(1 + \frac{1}{n}\right)^n$ 2

End of Question 7

(a) Consider the equation $z^7 - 1 = (z - 1)(z^6 + z^5 + z^4 + z^3 + z^2 + z + 1) = 0$

(i) Show that $v = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$ is a complex root of $z^7 - 1 = 0$ **1**

(ii) Show that the other five complex roots of $z^7 - 1 = 0$ are

$$v^k \text{ for } k = 2, 3, 4, 5, 6$$
2

(iii) Show that $\overline{(v^{7-k})} = v^k$ for $k = 1, 2, \dots, 6$

i.e. show that the conjugate of v^{7-k} is v^k **2**

(iv) Deduce that $v + v^2 + v^4$ and $v^3 + v^5 + v^6$ are conjugate complex numbers. **1**

(v) Deduce that $\cos \frac{\pi}{7} - \cos \frac{2\pi}{7} + \cos \frac{3\pi}{7} = \frac{1}{2}$ **3**

Question 8 is continued on the next page

- (b) (i) Use a suitable substitution to show that

$$\int_0^{\frac{\pi}{2}} \cos x \sin^{n-1} x \, dx = \frac{1}{n}, \quad n = 1, 2, 3, \dots \quad 1$$

- (ii) Show by integration that

$$\int x \sin x \, dx = -x \cos x + \sin x \quad 1$$

(iii) Let $t_n = \int_0^{\frac{\pi}{2}} x \sin^n x \, dx, \quad n = 0, 1, 2, \dots$

Use integration by parts to prove that

$$t_n = \frac{1}{n^2} + \frac{n-1}{n} t_{n-2}, \quad n = 2, 3, 4, \dots \quad 4$$

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Ques 1

$$(a) (i) \text{ Put } \frac{2}{1-x^2} = \frac{2}{(1-x)(1+x)} = \frac{A}{1-x} + \frac{B}{1+x}$$

$$: A(1+x) + B(1-x) = 2$$

$$\text{For } x=1, 2A=2, A=1 \Rightarrow B=1$$

$$\therefore \frac{2}{1-x^2} = \frac{1}{1-x} + \frac{1}{1+x}$$

$$(ii) \text{ From (i), } \int_0^{\frac{1}{4}} \frac{2}{1-x^2} dx = \int_0^{\frac{1}{4}} \frac{1}{1-x} + \frac{1}{1+x} dx$$

$$= [\ln(1+x) - \ln(1-x)]_0^{\frac{1}{4}}$$

$$= \ln \frac{5}{4} - \ln \frac{3}{4} = \ln \left(\frac{5}{3} \right)$$

$$(iii) \text{ Put } u = x^2 \quad ; \quad x=0, u=0$$

$$\frac{du}{dx} = 2x \quad x = \frac{1}{2}, u = \frac{1}{4}$$

$$\therefore I = \int_0^{\frac{1}{4}} \frac{du}{1-u^2} = \ln \left(\frac{5}{3} \right), \text{ from (ii)}$$

$$(b) \text{ Let } t = \tan x \quad x=0, t=0$$

$$\frac{dt}{dx} = \sec^2 x = 1+t^2 \quad x = \frac{\pi}{4}, t=1$$

$$\therefore I = \int_0^1 \frac{2 dt}{(1+t^2) \left[1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} \right]}$$

$$= \int_0^1 \frac{2 dt}{1+t^2+2t+1-t^2} = \int_0^1 \frac{1}{t+1} dt$$

$$= [\ln(t+1)]_0^1 = \ln 2$$

OR

$$I = \int_0^{\frac{\pi}{4}} \frac{2 dx}{2\cos^2 x + 2\sin x \cos x}$$

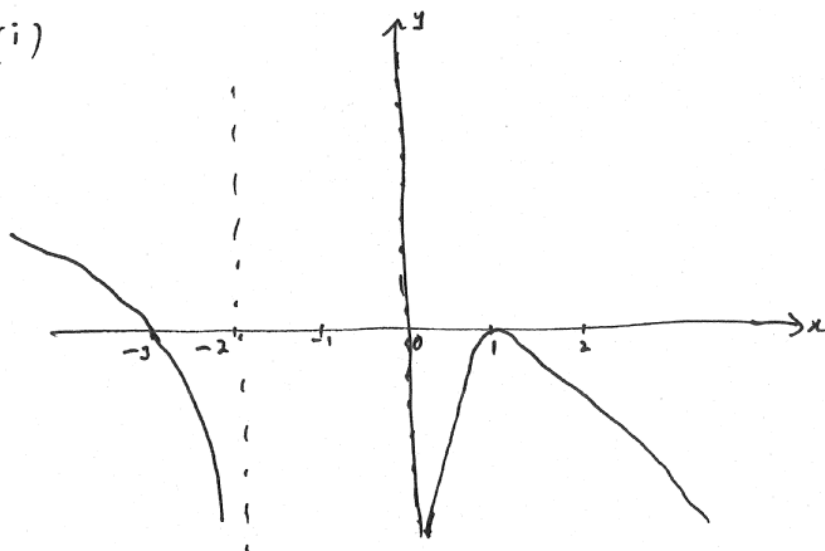
$$= \int_0^{\frac{\pi}{4}} \frac{\sec^2 x}{1 + \tan x} dx$$

$$= [\ln(1 + \tan x)]_0^{\frac{\pi}{4}}$$

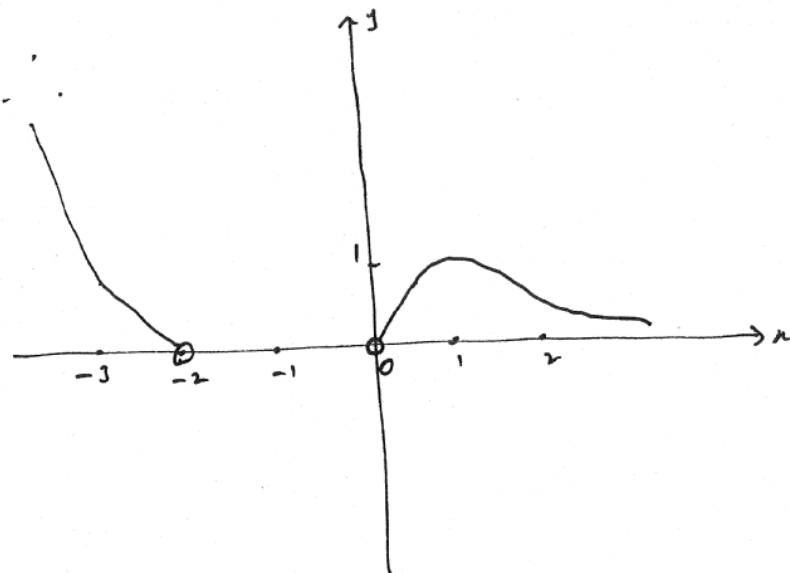
$$= \ln 2$$

$$\begin{aligned}
 (c) \quad I &= \int_0^1 \frac{4}{(2x+1)^2 + 4} dx = 4 \cdot \frac{1}{2} \left[\tan^{-1} \frac{2x+1}{2} \right]_0^1 \cdot \frac{1}{2} \\
 &= \tan^{-1} \frac{3}{2} - \tan^{-1} \frac{1}{2} \\
 &= \tan^{-1} \left(\frac{\frac{3}{2} - \frac{1}{2}}{1 + \frac{3}{2} \cdot \frac{1}{2}} \right) = \tan^{-1} \left(\frac{4}{7} \right)
 \end{aligned}$$

(d) (i)



(ii) $y = e^{\ln f(x)} = f(x)$ if $f(x) > 0$



Que 2

(a) (i) put $u = x-1$ $\frac{dv}{dx} = f'(x)$

$$\therefore \frac{du}{dx} = 1, \quad v = f(x)$$

$$\begin{aligned} \therefore \int_0^1 (x-1) f'(x) dx &= [(x-1) f(x)]_0^1 - \int_0^1 f(x) dx \\ &= 0 - (-f(0)) - \int_0^1 f(x) dx \\ &= f(0) - \int_0^1 f(x) dx \end{aligned}$$

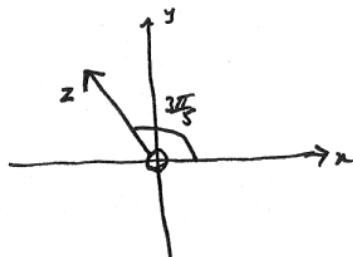
(ii) Hence $\dots \int_0^1 \frac{x-1}{(x+1)^2} dx \Rightarrow f'(x) = \frac{1}{(x+1)^2}, \quad f(x) = -\frac{1}{x+1}$

$$\begin{aligned} \therefore I &= -1 + \int_0^1 \frac{1}{x+1} dx = -1 + [\ln(x+1)]_0^1 \\ &= \ln 2 - 1 \end{aligned}$$

or, Otherwise $\dots$

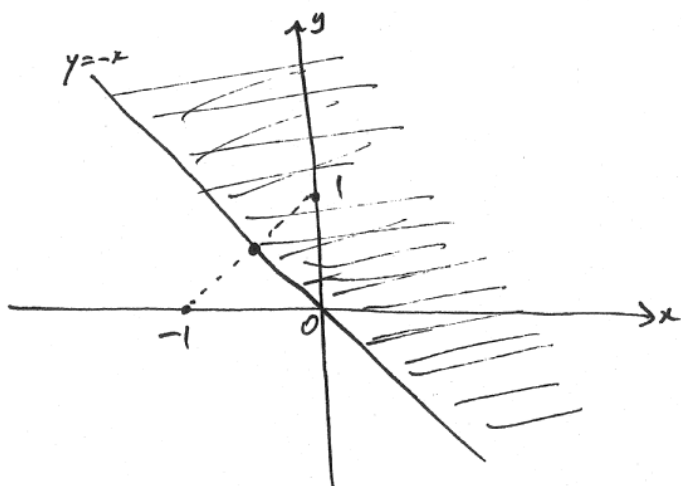
$$\begin{aligned} \int_0^1 \frac{x-1}{(x+1)^2} dx &= \int_0^1 \frac{x+1-2}{(x+1)^2} dx \\ &= \int_0^1 \frac{1}{x+1} - \frac{2}{(x+1)^2} dx \\ &= \left[\ln(x+1) + \frac{2}{x+1} \right]_0^1 \\ &= \ln 2 + 1 - (0 + 2) \\ &= \ln 2 - 1 \end{aligned}$$

(b) (i)

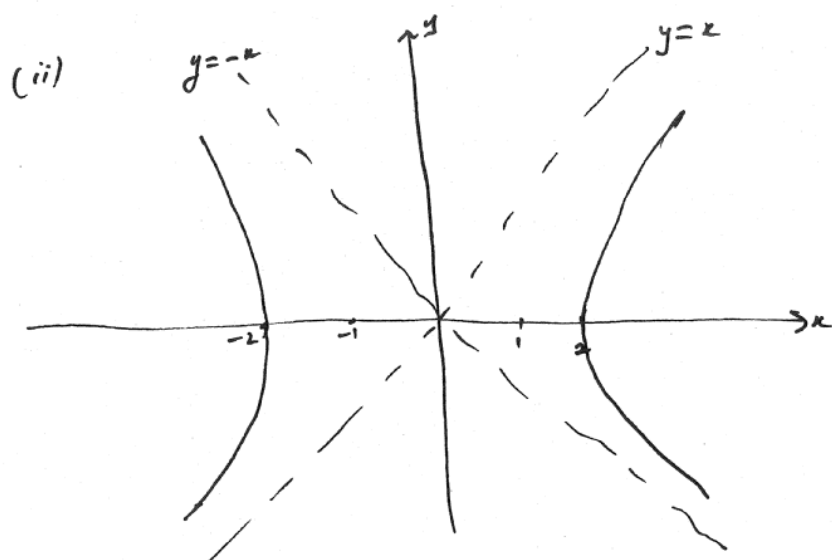


$$\begin{aligned} \text{(ii) } \arg(-2) &= \pi + \frac{3\pi}{5} \\ &= \frac{8\pi}{5} \quad \text{or} \quad -\frac{2\pi}{5} \end{aligned}$$

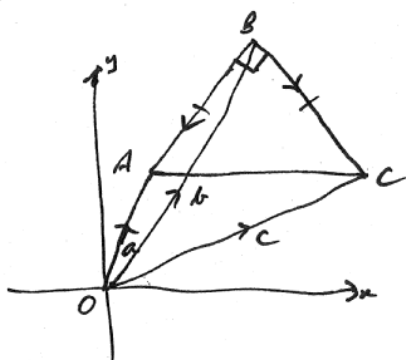
(c)



(d) (i) $\therefore (2x)^2 + (2iy)^2 = 4$
 $\Rightarrow x^2 - y^2 = 1$ [rectangular hyperbola]



(e) (i)



$$\vec{BA} = a - b$$

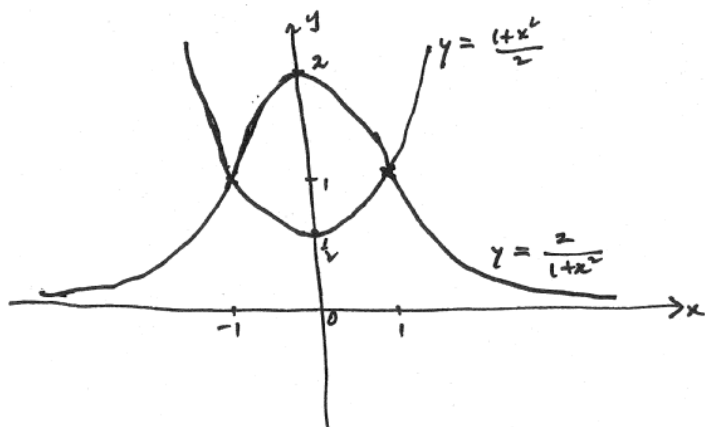
(ii) $\vec{BC} = i \vec{BA}$

$$\therefore c - b = i(a - b)$$

$$\therefore c = ai + b(1 - i)$$

Q. 3

(a) (i)



(ii) From (i), $0 < \frac{2}{1+x^2} \leq 2$

$$\therefore -1 < \frac{2}{1+x^2} - 1 \leq 1$$

i.e. range is $-1 < y \leq 1$

(b) (i) $\frac{2}{1+x^2} - 1 = \frac{2 - (1+x^2)}{1+x^2} = \frac{1-x^2}{1+x^2}$

$\therefore$ from (a)(ii), range is $0 \leq y < \pi$

(ii) $\frac{d}{dx} \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = -\frac{1}{\sqrt{1 - \left(\frac{1-x^2}{1+x^2}\right)^2}} \cdot -2(1+x^2)^{-2} \cdot 2x$

$$= \frac{4x}{\sqrt{(1+x^2)^2 - (1-x^2)^2} \cdot (1+x^2)}$$

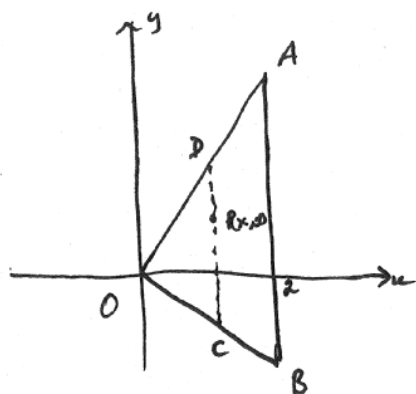
$$= \frac{4x}{(1+x^2)\sqrt{4x^2}} = \frac{2x}{(1+x^2)\sqrt{x^2}}$$

$\therefore$ (d), if $x > 0$, $\frac{dy}{dx} = \frac{2}{1+x^2}$

& (e), if $x < 0$, $\frac{dy}{dx} = \frac{-2}{1+x^2}$

Q. 4

(a)



$$\Rightarrow CD = 2x + x = 3x$$

$$\therefore \delta V \approx 3x(4x^2 + x) \delta x$$

$$\begin{aligned} \therefore V &= \int_0^2 12x^3 + 3x^2 dx \\ &= [3x^4 + x^3]_0^2 = 56 \text{ u}^2 \end{aligned}$$

(b) $\therefore x^2 y \frac{dy}{dx} + y^2 = 2x$

$$\Rightarrow \frac{dy}{dx} = \frac{2x - y^2}{x^2 y} = \frac{1}{y} - \frac{y}{2x}$$

(c) (i)

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

$$= -\frac{b^2}{a^2} \cdot \frac{a \cos \theta}{b \sin \theta} \text{ at } P$$

$$= -\frac{b \cos \theta}{a \sin \theta}$$

$$\therefore \text{tangent at } P \text{ is } y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$\therefore \frac{\sin \theta}{b} y - \sin^2 \theta = -\frac{\cos \theta}{a} x + \cos^2 \theta$$

$$\therefore \frac{\cos \theta}{a} x + \frac{\sin \theta}{b} y = \cos^2 \theta + \sin^2 \theta = 1$$

(ii) Rewrite as $\frac{\cos \lambda}{k} x + \frac{\sin \lambda}{k} y = 1$

Now, from (ii), we need $\frac{\cos \lambda}{k} = \frac{\cos \theta}{a}$ and $\frac{\sin \lambda}{k} = \frac{\sin \theta}{b}$

$$\Rightarrow (a \cos \lambda)^2 + (b \sin \lambda)^2 = k^2 \cos^2 \theta + k^2 \sin^2 \theta$$

$$= k^2 (\cos^2 \theta + \sin^2 \theta)$$

i.e. $a^2 \cos^2 \lambda + b^2 \sin^2 \lambda = k^2$

(iii)

From (ii) & (iii),

$$AQ^2 + BR^2 = \frac{(a \cos \lambda - k)^2 + (a \sin \lambda + k)^2}{\cos^2 \lambda + \sin^2 \lambda}$$

$$= 2(a^2 \cos^2 \lambda + k^2)$$

$$= 2((a^2 - b^2) \sin^2 \lambda + a^2 \cos^2 \lambda + b^2 \sin^2 \lambda), \text{ from (i), (iii)}$$

$$= 2(a^2 (\sin^2 \lambda + \cos^2 \lambda))$$

$$= 2a^2$$

Qu 5

(a) (i) $m\ddot{x} = mg - mkv^2$

$$\Rightarrow \ddot{x} = g - kv^2 \Rightarrow g - kV^2 = 0 \quad \text{or} \quad V^2 = \frac{g}{k}$$

$$\therefore \ddot{x} = k \left(\frac{g}{k} - v^2 \right) = k(V^2 - v^2)$$

(ii) $m\ddot{x} = -mg - mkv^2$

↑
x=0

$$\therefore \ddot{x} = -k(V^2 + v^2)$$

$$\therefore v \frac{dv}{dx} = -k(V^2 + v^2)$$

$$\text{or} \quad \frac{dx}{dv} = -\frac{1}{k} \cdot \frac{v}{V^2 + v^2}$$

$$\therefore \text{max ht} = -\frac{1}{k} \int_{V^2}^0 \frac{v}{V^2 + v^2} dv$$

$$= -\frac{1}{2k} \left[\ln(V^2 + v^2) \right]_{V^2}^0$$

$$= \frac{1}{2k} \left(\ln(V^2 + v^4) - \ln V^2 \right)$$

$$= \frac{1}{2k} \ln(1 + v^2)$$

(iii) $v \frac{dv}{dx} = k(V^2 - v^2) \Rightarrow \frac{dx}{dv} = \frac{1}{k} \cdot \frac{v}{V^2 - v^2}$

$$\therefore \text{from (ii)}, \quad \frac{1}{2k} \ln(1 + v^2) = \frac{1}{k} \int_0^v \frac{v}{V^2 - v^2} dv$$

$$= -\frac{1}{2k} \left[\ln(V^2 - v^2) \right]_0^v$$

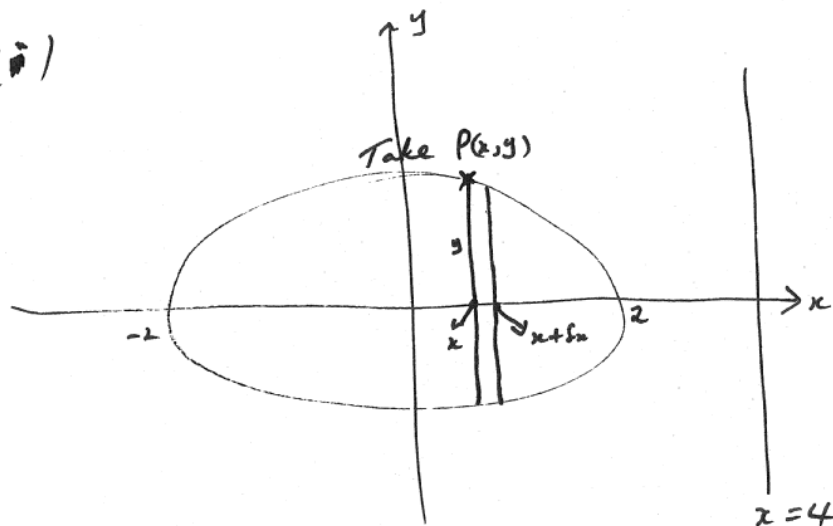
$$= \frac{1}{2k} \left(\ln V^2 - \ln(V^2 - v^2) \right) = \frac{1}{2k} \ln \left(\frac{V^2}{V^2 - v^2} \right)$$

$$\Rightarrow \frac{V^2}{V^2 - v^2} = 1 + v^2 \quad \text{or} \quad V^2 - v^2 = \frac{V^2}{1 + v^2}$$

$$\text{or} \quad v^2 = V^2 - \frac{V^2}{1 + v^2} = \frac{V^4}{1 + v^2}$$

$$\therefore v^2 = \frac{1 + v^2}{V^4} = V^{-2} + V^{-4}$$

b (ii)



$$\therefore \delta V \approx \pi [(4-x)^2 - (4-x-\delta x)^2] 2y$$

$$\approx 2\pi [2(4-x)\delta x]y, \text{ ignoring } \delta x^2 \text{ term}$$

$$= 4\pi (4-x)y \delta x \quad : y^2 = 3\left(1 - \frac{x^2}{4}\right) = \frac{3}{4}(4-x^2)$$

$$\therefore V = 4\pi \int_{-2}^2 (4-x) \frac{\sqrt{3}}{2} \sqrt{4-x^2} dx$$

$$= 2\pi\sqrt{3} \int_{-2}^2 (4-x) \sqrt{4-x^2} dx$$

$$= 8\sqrt{3}\pi \int_{-2}^2 \sqrt{4-x^2} dx - 2\sqrt{3}\pi \int_{-2}^2 x \sqrt{4-x^2} dx$$

$$(ii) V = 8\sqrt{3}\pi \int_{-2}^2 \sqrt{4-x^2} dx \quad \text{since } x \sqrt{4-x^2} \text{ is an odd function}$$

$$= 8\sqrt{3}\pi \cdot \frac{1}{2} \pi \cdot 2^2 \quad \text{[semi-circle]}$$

$$= 16\sqrt{3}\pi^2$$

Qn 6

$$(a) (i) \frac{dy}{dx} = \frac{3}{2} x^{\frac{1}{2}} = \frac{3}{2} t \text{ at } P$$

$$\therefore \text{tangent at } P \text{ is } y - t^3 = \frac{3}{2} t (x - t^2)$$

$$\text{or } 3tx - 2y + 2t^3 - 3t^3 = 0$$

$$\text{i.e. } 3tx - 2y - t^3 = 0$$

(ii) The tangent at $P(x^1, t^1)$ is a cubic equation in t

$$\Rightarrow \text{at most 3 values for } t \text{ for } 3tx_1 - 2y_1 - t^3 = 0$$

$$\Rightarrow \text{at most 3 tangents}$$

(iii) Now, $t^3 - 3x_1 t + 2y_1 = 0$ has roots t_1, t_2, t_3

$$\therefore (a) \sum t_i^3 - 3x_1 \sum t_i + 3(2y_1) = 0 \text{ where } \sum t_i = 0$$

$$\therefore \sum t_i^3 = -6y_1$$

$$(b) \sum (t_i t_j)^2 = (t_1 t_2 + t_2 t_3 + t_3 t_1)^2 - 2(t_1 t_2 t_3 t_1 + t_2 t_3 t_1 t_2 + t_3 t_1 t_2 t_3)$$

$$= (-3x_1)^2 - 2t_1 t_2 t_3 (t_1 + t_2 + t_3)$$

$$= 9x_1^2 \text{ since } \sum t_i = 0$$

$$(iv) \text{ It is } \left(\frac{1}{t}\right)^3 - 3x_1 \left(\frac{1}{t}\right) + 2y_1 = 0$$

$$\text{i.e. } 2y_1 t^3 - 3x_1 t^2 + 1 = 0$$

$$(b) (i) \text{ put } x+y = A \quad \Rightarrow 2x = A+c$$

$$x-y = c \quad \Rightarrow 2y = A-c$$

$$\therefore \sin A + \sin C = 2 \sin \frac{A+C}{2} \cos \frac{A-C}{2} \quad \text{from data}$$

$$(ii) \frac{\sin A}{2a} = \frac{\sin B}{a+c} = \frac{\sin C}{2c}$$

$$\therefore \sin A + \sin C = \left(\frac{2a}{a+c} + \frac{2c}{a+c} \right) \sin B$$

$$= \frac{2(a+c)}{a+c} \sin B = 2 \sin B$$

$$(B) 2 \sin B = 2 \sin \frac{A+C}{2} \cos \frac{A-C}{2} \quad \text{from (i)}$$

$$\Rightarrow 2 \sin \frac{B}{2} \cos \frac{B}{2} = \sin \frac{A+C}{2} \cos \frac{A-C}{2}$$

$$= \sin \left(\frac{\pi}{2} - \frac{B}{2} \right) \cos \frac{A-C}{2} \quad \text{since } A+B+C = \pi$$

$$= \cos \frac{B}{2} \cos \frac{A-C}{2}$$

$$\therefore \sin \frac{B}{2} = \frac{1}{2} \cos \frac{A-C}{2}$$

Q. 7

$$\begin{aligned}
 (i) \quad f(n+1) - f(n) &= (n+2)^3 + \dots + (2n)^3 + (2n+1)^3 + (2n+2)^3 - ((n+1)^3 + \dots + (2n)^3) \\
 &= (2n+1)^3 + (2n+2)^3 - (n+1)^3 \\
 &= (2n+1)^3 + 8(n+1)^3 - (n+1)^3 \\
 &= (2n+1)^3 + 7(n+1)^3
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \Delta S &= \frac{2n+1}{4} (4(2n+1)^2 - (3n+1)(5n+3)) \\
 &= \frac{2n+1}{4} (16n^2 + 16n + 4 - 15n^2 - 14n - 3) \\
 &= \frac{2n+1}{4} (n^2 + 2n + 1) = \frac{2n+1}{4} (n+1)^2
 \end{aligned}$$

$$(iii) \quad f(1) = 2^3 = 8 \quad \text{and} \quad \frac{1^2}{4} (4)(8) = 8$$

$\therefore$ Assume $f(n) = (n+1)^3 + \dots + (2n)^3 = \frac{n^2}{4} (3n+1)(5n+3)$ for some integer $n \geq 1$

Then, $f(n+1) = f(n) + (2n+1)^3 + 7(n+1)^3$ from (i)

$$= \frac{n^2}{4} (3n+1)(5n+3) + (2n+1)^3 + 7(n+1)^3, \text{ using the assumption}$$

$$= \frac{(n+1)^2}{4} (3n+1)(5n+3) - \frac{2n+1}{4} (3n+1)(5n+3) + (2n+1)^3 + 7(n+1)^3$$

$$= \frac{(n+1)^2}{4} (3n+1)(5n+3) + \frac{2n+1}{4} (n+1)^2 + 7(n+1)^3, \text{ from (ii)}$$

$$= \frac{(n+1)^2}{4} [(3n+1)(5n+3) + 2n+1 + 28(n+1)]$$

$$= \frac{(n+1)^2}{4} (15n^2 + 44n + 32)$$

$$= \frac{(n+1)^2}{4} (3n+4)(5n+8) = \frac{(n+1)^2}{4} (3(n+1)+1)(5(n+1)+3)$$

$\therefore$ if the result is true for n it is also true for $n+1$.

But it is correct for $n=1$

$\therefore$ by induction, $(n+1)^3 + \dots + (2n)^3 = \frac{n^2}{4} (3n+1)(5n+3) \quad \forall n \geq 1$

$$\begin{aligned}
 \text{(iv)} \quad (n+1)^3 + \dots + (2n)^3 &= 1^3 + \dots + n^3 + \dots + (2n)^3 - (1^3 + \dots + n^3) \\
 &= \left(\frac{2n}{2} (2n+1) \right)^2 - \left(\frac{n}{2} (n+1) \right)^2 \\
 &= \frac{n^2}{4} \left(4(2n+1)^2 - (n+1)^2 \right) \\
 &= \frac{n^2}{4} (4n+2-n-1)(4n+2+n+1) \\
 &= \frac{n^2}{4} (3n+1)(5n+3)
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad \text{(i)} \quad \frac{\binom{n}{k}}{n^k} &= \frac{n!}{(n-k)! k! n^k} \\
 &= \frac{n(n-1)(n-2) \dots (n-k+1)}{k! n^k} \\
 &= \frac{1 \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{k-1}{n}\right)}{k!}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \text{From (i)}, \quad \frac{\binom{n+1}{k}}{(n+1)^k} &= \frac{\left(1 - \frac{1}{n+1}\right) \left(1 - \frac{2}{n+1}\right) \dots \left(1 - \frac{k-1}{n+1}\right)}{k!} \\
 &> \frac{\left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{k-1}{n}\right)}{k!} \quad \text{since } \frac{1}{n+1} < \frac{1}{n} \\
 &= \frac{\binom{n}{k}}{n^k}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad \left(1 + \frac{1}{n+1}\right)^{n+1} &= 1 + (n+1) \frac{1}{n+1} + \frac{\binom{n+1}{2}}{(n+1)^2} + \dots + \frac{\binom{n+1}{k}}{(n+1)^k} + \dots + \frac{\binom{n+1}{n}}{(n+1)^n} + \frac{1}{(n+1)^{n+1}} \\
 &> \left(1 + 1 + \frac{\binom{n}{2}}{n^2} + \dots + \frac{\binom{n}{k}}{n^k} + \dots + \frac{\binom{n}{n}}{n^n}\right) + \frac{1}{(n+1)^{n+1}} \\
 &> 1 + 1 + \dots + \frac{\binom{n}{k}}{n^k} + \dots + \frac{\binom{n}{n}}{n^n} \\
 &= \left(1 + \frac{1}{n}\right)^n
 \end{aligned}$$

Qn 8

$$(a) \quad (i) \quad v^7 - 1 = \left(\cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} \right)^7 - 1$$
$$= \cos 2\pi + i \sin 2\pi - 1 = 1 - 1 = 0$$

ie v is a root of $z^7 - 1 = 0$

$$(ii) \quad v^k = \cos \frac{2\pi k}{7} + i \sin \frac{2\pi k}{7}$$

$$\therefore (v^k)^7 = \cos 2\pi k + i \sin 2\pi k$$
$$= 1 \text{ if } k=2, 3, \dots, 6$$

$\Rightarrow v^2, v^3, \dots, v^6$ are also roots of $z^7 = 1$

$$(iii) \quad \overline{(v^{7-k})} = \cos \frac{2\pi(7-k)}{7} - i \sin \frac{2\pi}{7}(7-k)$$
$$= \cos \left(-\frac{2\pi k}{7} \right) - i \sin \left(-\frac{2\pi k}{7} \right)$$
$$= \cos \frac{2\pi k}{7} + i \sin \frac{2\pi k}{7} = v^k$$

$$(iv) \quad \overline{v + v^2 + v^4} = \bar{v} + \bar{v}^2 + \bar{v}^4$$
$$= v^6 + v^5 + v^3 \text{ from (iii)}$$

ie $v + v^2 + v^4$ and $v^3 + v^5 + v^6$ are conjugates

$$(v) \quad \text{From (iv), } (v + v^2 + v^4) + (v^3 + v^5 + v^6)$$

$$= 2 \operatorname{Re}(v + v^2 + v^4)$$

$$= 2 \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} \right)$$

$$= 2 \left(\cos \frac{2\pi}{7} - \cos \frac{3\pi}{7} - \cos \frac{\pi}{7} \right)$$

$$= -1 \text{ since } v^6 + v^5 + \dots + v^2 + v + 1 = 0$$

$$\therefore \cos \frac{\pi}{7} - \cos \frac{2\pi}{7} + \cos \frac{3\pi}{7} = \frac{1}{2}$$

(4) (i) put $u = \sin x$; $x=0, u=0$
 $\frac{du}{dx} = \cos x$ $x = \frac{\pi}{2}, u=1$

$$\therefore I = \int_0^1 u^{n-1} du = \left[\frac{u^n}{n} \right]_0^1 = \frac{1}{n}$$

(ii) put $u = x$, $\frac{dv}{dx} = \sin x$

$$\therefore \frac{du}{dx} = 1, \quad v = -\cos x$$

$$\therefore \int x \sin x dx = x(-\cos x) - \int (-\cos x) dx$$

$$= -x \cos x + \sin x$$

(iii) As suggested from (ii),

put $u = \sin^{n-1} x$, $\frac{dv}{dx} = x \sin x$

$$\therefore \frac{du}{dx} = (n-1) \sin^{n-2} x \cos x, \quad v = -x \cos x + \sin x$$

$$\therefore t_n = \left[\sin^{n-1} x (-x \cos x + \sin x) \right]_0^{\frac{\pi}{2}} - (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos x (-x \cos x + \sin x) dx$$

$$= 1 - (n-1) \int_0^{\frac{\pi}{2}} -x \sin^{n-2} x (1 - \sin^2 x) + \cos x \sin^{n-1} x dx$$

$$= 1 + (n-1) t_{n-2} - (n-1) t_n - (n-1) \int_0^{\frac{\pi}{2}} \cos x \sin^{n-1} x dx$$

$$= 1 + (n-1) t_{n-2} - (n-1) t_n - \frac{n-1}{n}, \text{ from (i)}$$

$$\therefore n t_n = 1 - 1 + \frac{1}{n} + (n-1) t_{n-2}$$

$$\text{or, } t_n = \frac{1}{n} + \frac{n-1}{n} t_{n-2}, \quad n=2, 3, \dots$$